

Steel Industry Energy Consumption Prediction

Week 2 Internship Project Documentation

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Project Overview

Objective

The objective of this project was to analyze energy consumption patterns in a steel manufacturing plant and develop baseline machine learning models capable of predicting electricity usage.

The project followed a complete machine learning workflow, beginning with Exploratory Data Analysis (EDA) and Feature Engineering, followed by Data Preprocessing, Regression Modeling, Model Evaluation, and Model Selection.

This project provided practical experience in understanding real-world industrial data and applying machine learning techniques to solve a regression problem.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the Steel Industry Energy Consumption dataset before building predictive models. The objective was to examine the dataset structure, assess data quality, identify relationships between variables, and uncover patterns that could influence energy consumption. This analysis also guided the feature engineering process and helped ensure that the dataset was suitable for machine learning.

Dataset Inspection

The dataset was first explored by examining its dimensions, feature names, data types, and descriptive statistics. It contains **35,040 observations** and **11 original features**,

representing electricity consumption measurements recorded every **15 minutes** throughout the year **2018**. This initial inspection provided a clear understanding of the available information and confirmed that the dataset contained both numerical and categorical variables relevant to energy consumption.

Data Quality Assessment

A data quality assessment was performed to identify missing values, incorrect data types, and inconsistencies that could affect model performance.

The original dataset contained **no missing values**, indicating that it was complete and well-maintained. During feature engineering, one missing value was introduced while calculating the **Power Factor Ratio** because a single observation had a lagging power factor equal to zero, making the division undefined. Since this represented only one record out of more than thirty-five thousand observations, its impact on the analysis was negligible.

Overall, the dataset demonstrated high data quality with no significant issues requiring extensive preprocessing.

Feature Engineering

To enrich the dataset and provide additional information for predictive modeling, several new features were created.

The **date** column was converted into a datetime format, allowing meaningful time-based attributes to be extracted. New features including **Hour**, **Day of Week**, **Month**, and **Day Type (Weekday/Weekend)** were generated to capture temporal variations in electricity usage.

A new feature named **Power_Factor_Ratio** was created by dividing the Leading Current Power Factor by the Lagging Current Power Factor. This ratio represents the relationship between the two power factors and provides additional insight into the plant's electrical operating conditions.

Additionally, a binary feature named **High_Load** was created using the **75th percentile** of the target variable (**Usage_kWh**). Records with energy consumption above this threshold were labeled as high-load observations. This feature was created solely for exploratory analysis and was intentionally excluded from model training to prevent target leakage.

Outlier Analysis

Outliers in the **Usage_kWh** variable were detected using the **Interquartile Range (IQR)** method.

The analysis identified **328 outliers**, representing approximately **0.94%** of the dataset. A boxplot was used to visualize these observations and understand their distribution.

Rather than removing these values, they were retained because they correspond to genuine periods of exceptionally high electricity demand during plant operations. Removing them would have eliminated meaningful operational behavior and potentially reduced the model's ability to learn real-world energy consumption patterns.

Correlation Analysis

A correlation heatmap was generated to measure the linear relationship between numerical variables and identify the features most strongly associated with energy consumption.

The analysis showed that **CO₂ emissions** exhibited the strongest positive correlation with **Usage_kWh**, followed by **Lagging Current Reactive Power** and **Lagging Current Power Factor**. These findings indicate that increases in electricity consumption are closely associated with higher CO₂ emissions and greater reactive power demand.

Understanding these relationships helped identify the most informative variables for regression modeling.

Load Type Analysis

Average energy consumption was compared across the three operational load categories: **Light Load**, **Medium Load**, and **Maximum Load**.

The grouped bar chart revealed a clear increase in average electricity consumption as operational load increased. **Maximum Load** recorded the highest average energy usage, followed by **Medium Load**, while **Light Load** showed the lowest consumption levels.

This confirms that the operational load type is an important indicator of the plant's energy demand.

Hourly Energy Consumption Analysis

The hourly trend of average electricity consumption was analyzed to understand how energy usage changes throughout the day.

The analysis showed that electricity consumption gradually increases during working hours, reaches its highest average around **9:00 AM**, and then gradually decreases later in the day. Lower consumption levels were observed during late-night and early-morning hours, reflecting reduced production activity outside normal operating periods.

These temporal patterns demonstrate that production schedules have a significant influence on electricity demand.

Key EDA Findings

The exploratory analysis provided several important insights into the dataset:

- The dataset was clean, complete, and suitable for machine learning with minimal preprocessing.
- Time-based feature engineering introduced valuable information about daily and seasonal energy consumption patterns.
- Only **0.94%** of observations were identified as outliers, and these represented genuine high-demand operating conditions rather than data errors.
- **CO₂ emissions**, **Lagging Current Reactive Power**, and **Lagging Current Power Factor** were identified as the strongest predictors of electricity consumption.
- Energy usage increased significantly during **Maximum Load** operations and followed a clear hourly production cycle, with peak consumption occurring during active working hours.

These findings established a strong foundation for developing baseline regression models and selecting meaningful predictive features.

Model Training Process

Following the completion of exploratory analysis and feature engineering, the dataset was prepared for predictive modeling. The objective of this stage was to develop baseline regression models capable of accurately estimating electricity consumption while comparing the strengths and limitations of different machine learning algorithms.

Data Preparation

Before training the models, the dataset underwent additional preprocessing to ensure that only relevant information was provided to the learning algorithms.

The original **date** column was removed because its information had already been captured through engineered time-based features. The **High_Load** feature was also excluded because it was directly derived from the target variable (**Usage_kWh**) and would introduce target leakage if used during training. Duplicate categorical information, such as **WeekStatus** and **Day_of_week**, was removed to eliminate redundancy after feature engineering.

Categorical variables were transformed using **One-Hot Encoding**, allowing machine learning models to interpret non-numerical information without introducing ordinal relationships between categories.

Finally, the dataset was divided into **80% training data** and **20% testing data** using a fixed random state of **42** to ensure reproducibility.

Model Development

Four baseline regression algorithms were implemented and trained using the prepared dataset:

- Linear Regression
- Ridge Regression
- Decision Tree Regressor
- Random Forest Regressor

These models represent both linear and tree-based approaches, providing a comprehensive comparison of different regression techniques on the same dataset.

Model Evaluation

Each model was evaluated using multiple performance metrics to assess prediction accuracy and generalization capability.

The following evaluation metrics were used:

- **Mean Absolute Error (MAE)** to measure the average prediction error.
- **Root Mean Squared Error (RMSE)** to penalize larger prediction errors.
- **R² Score** to measure how well the model explains the variability in energy consumption.
- **5-Fold Cross Validation** to evaluate model stability across different subsets of the dataset.

Using multiple evaluation metrics provides a more reliable comparison than relying on a single performance measure.

Model Comparison

The comparison showed clear differences in predictive performance among the four regression models.

The **Random Forest Regressor** achieved the lowest MAE and RMSE, the highest R^2 Score (**0.999**), and the best average Cross Validation RMSE (**2.103**). The **Decision Tree Regressor** also performed well but produced slightly higher prediction errors than Random Forest. Linear Regression provided a strong baseline despite its assumption of linear relationships, while Ridge Regression achieved the lowest overall performance on this dataset.

These results indicate that the relationship between the input features and electricity consumption is largely non-linear, making ensemble tree-based methods more suitable than traditional linear regression models.

Results and Conclusion

The project successfully demonstrated the complete workflow of a supervised machine learning regression problem using real-world industrial energy consumption data.

The exploratory analysis provided valuable insights into the dataset, highlighting the importance of operational load, reactive power, and CO₂ emissions in influencing electricity consumption. Feature engineering enhanced the dataset by introducing meaningful temporal and electrical features that improved its predictive capability.

Among the evaluated algorithms, the **Random Forest Regressor** consistently achieved the best overall performance, producing highly accurate predictions and the strongest generalization ability across cross-validation. Its superior performance indicates that ensemble learning techniques are well suited for capturing the complex relationships present in industrial energy consumption data.

Overall, this project reinforced the importance of understanding data before modeling, applying appropriate preprocessing techniques, engineering meaningful features, and evaluating models using multiple performance metrics. These practices form the foundation of building reliable and interpretable machine learning solutions for real-world applications.